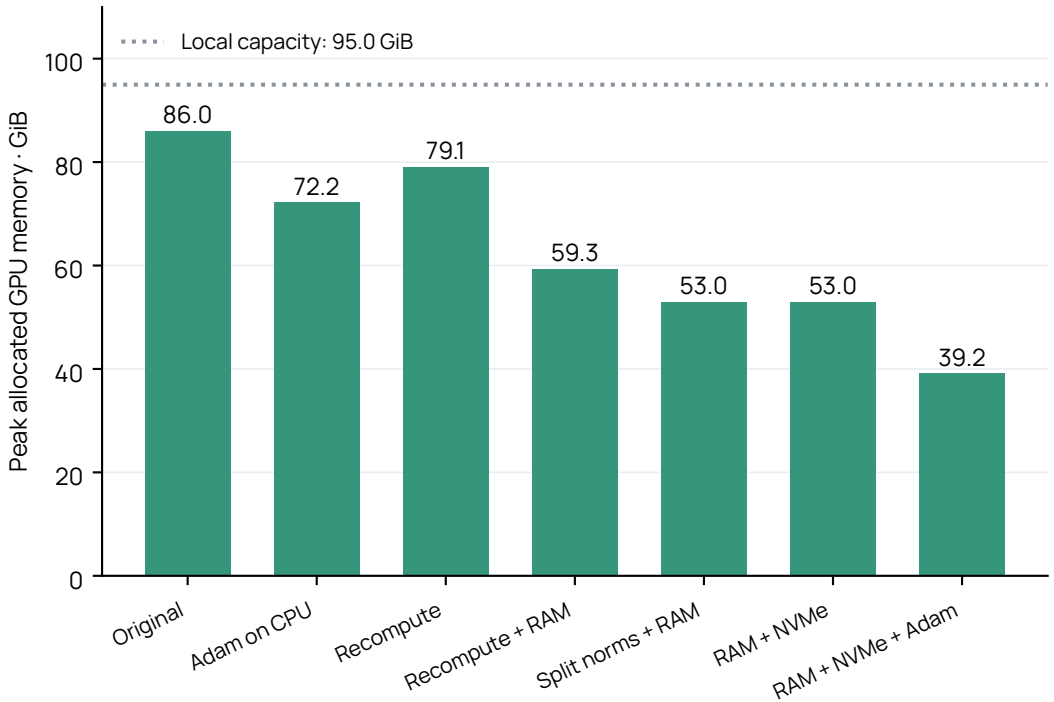
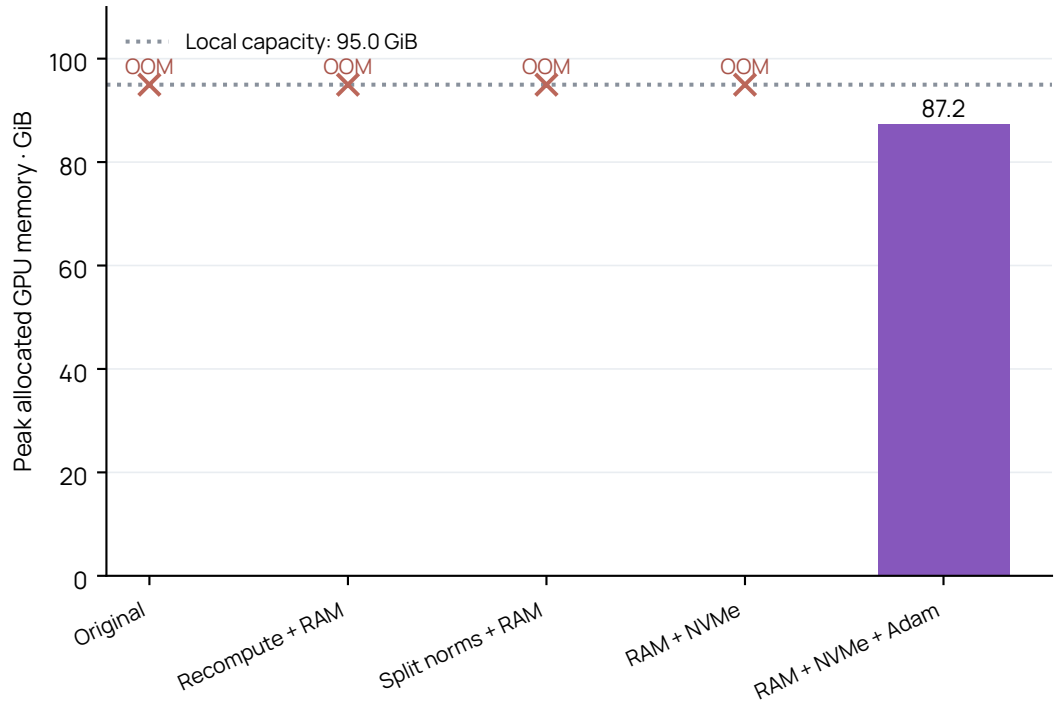
Local memory diagnostics · real step-2,000 checkpoint · batch 128 · microbatch 32

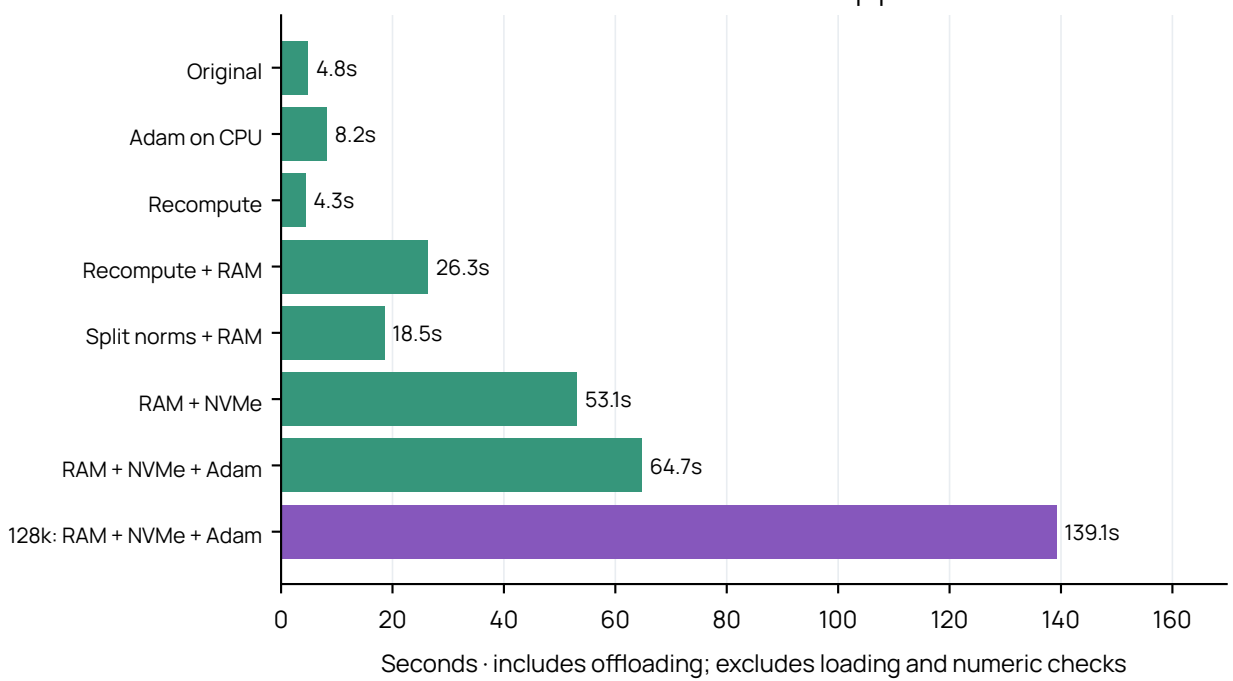
43k native vocabulary · complete training step



128k native vocabulary · complete training step



Runtime · one measured step per variant



43k correctness · full outputs + all 349 parameter gradients

Variant	Output exact	Gradient ΔL2	RNG exact
Original repeated	Yes	1.4e-08	Yes
Adam on CPU	Yes	1.4e-08	Yes
Recompute	Yes	1.4e-08	Yes
Recompute + RAM	Yes	1.4e-08	Yes
Split norms + RAM	Yes	1.3e-08	Yes
RAM + NVMe	Yes	1.3e-08	Yes
RAM + NVMe + Adam	Yes	1.4e-08	Yes

128k forward: separate full-table check. No original 128k gradient comparison.
All 26 attempts, including superseded wrappers and failures: results.json